

LECTURE 2

1 Hour

View of Data & Database Languages

Data Abstraction, Independence, DDL and DML

Our TextBooks:

- 1** “Database System Concepts”, 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill.
- 2** “Principles of Database and Knowledge – Base Systems”, Vol 1 by J. D. Ullman, Computer Science Press.
- 3** “Fundamentals of Database Systems”, 5th Edition by R. Elmasri and S. Navathe, Pearson Education.

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Learning Objectives

By the end of this session, students will be able to:

- 1 Describe the three levels of data abstraction
- 2 Distinguish between a database instance and a schema
- 3 Explain physical and logical data independence
- 4 Differentiate between DDL and DML

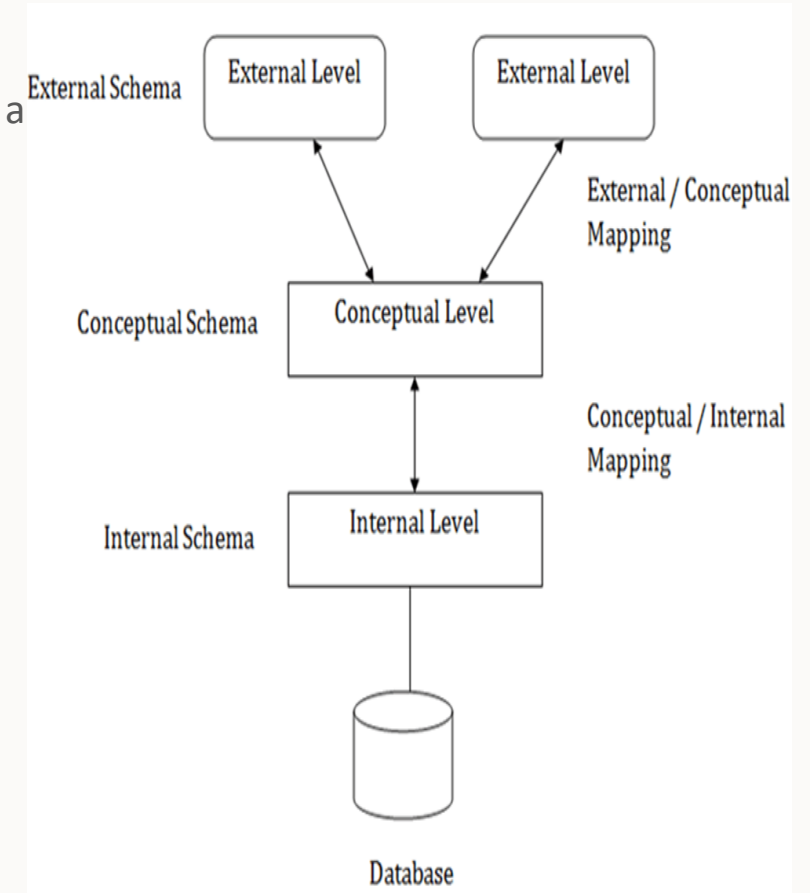
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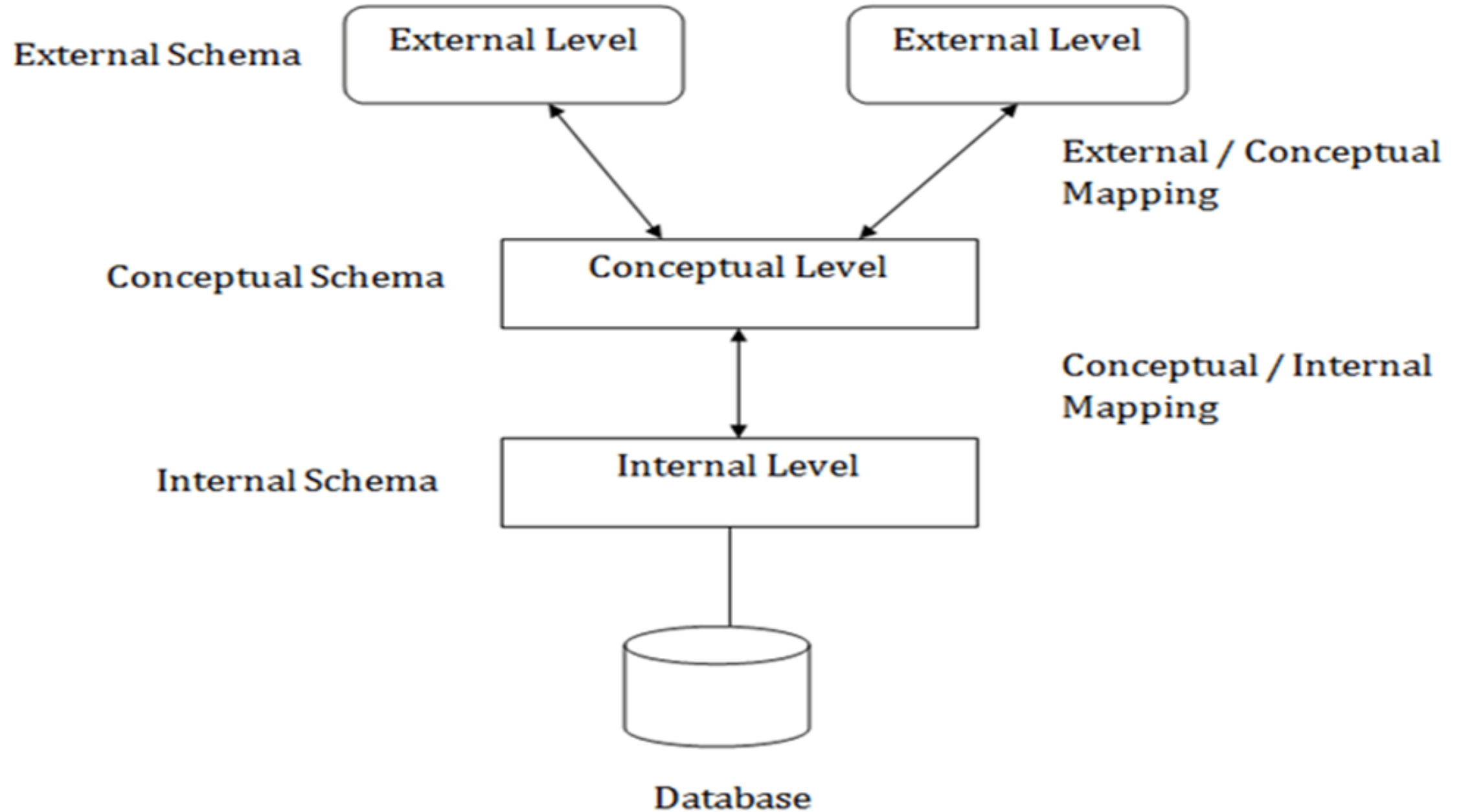
View of Data: Levels of Abstraction

How a DBMS hides complexity from users

- **View level - External Schema:** Describes only the part of the database relevant to a specific user
- **Logical level – Conceptual Schema:** Describes what data is stored and the relationships among it
- **Physical level – Internal Schema:** Describes how data is actually stored (files, indexes, blocks)

Purpose: Each higher level hides the complexity of the level below it





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Instances and Schemas

Two related but distinct concepts

- **Schema:** The overall design/structure of the database — changes rarely
- **Instance:** The actual data stored in the database at a given moment — changes constantly
- **Analogy:** Schema is like a blueprint; instance is like the furnished house at one point in time
- **Schema types:** Physical schema, logical schema, and view/subschema, matching the abstraction levels

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Data Independence

Isolating applications from storage/structure changes

Physical Data Independence

- Ability to change the physical storage/organization
- Without altering the logical schema
- E.g., changing file structure or adding indexes

Logical Data Independence

- Ability to change the logical schema
- Without rewriting application programs
- E.g., adding a new attribute to a relation

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Database Languages: Overview

How users and programs communicate with a DBMS

- **DDL:** Data Definition Language — defines and modifies database schema/structure
- **DML:** Data Manipulation Language — retrieves and modifies stored data
- **DCL:** Data Control Language — manages permissions (GRANT, REVOKE)
- **TCL:** Transaction Control Language — manages transactions (COMMIT, ROLLBACK)

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DDL & DML: Commands at a Glance

Common SQL commands by category

Category	Command	Purpose
DDL	CREATE	Create a new table, view, or database object
DDL	ALTER	Modify the structure of an existing object
DDL	DROP	Permanently remove a database object
DML	SELECT	Retrieve data from one or more tables
DML	INSERT	Add new rows of data into a table
DML	UPDATE	Modify existing data in a table
DML	DELETE	Remove existing rows from a table

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Recap & Quick Check

View of Data & Database Languages

Key Takeaways

- Three abstraction levels: physical, logical, and view
- Schema = structure (stable); Instance = data (changes often)
- Physical independence isolates storage changes; logical independence isolates schema changes
- DDL defines structure; DML manipulates data

Quick Check

Q1. What is the difference between a schema and an instance?

Q2. Give one example each of physical and logical data independence.

Q3. Which SQL command belongs to DDL: ALTER or UPDATE?